3 CPU SCHEDULING

3.1 Scheduling Criteria

CPU scheduling will occur when:

- A new process arrives into the Ready state.
- A process undergoes Wait → Ready state transition.
- A process undergoes Run → Wait state transition for an I/O request.
- A process undergoes Run → Ready state transition every q second where q is the time slice.
- The priority of a ready process is higher than the priority of a running process.

3.2 Goals of CPU Scheduling

- Maximize CPU utilization.
- Minimize the response time and waiting time of the processes.

3.3 Process times

3.3.1 Arrival Time (AT)

The time at which the process arrives into the Ready state is called arrival time of the process.

3.3.2 Burst Time (BT)

The time required by the process for its complete execution is called burst time/service time of the process.

3.3.3 Completion Time (CT)

The time by which a process completes its execution post arrival is called completion time of the process.

3.3.4 Turnaround Time (TAT)

The time difference between completion and arrival times of a process is called as the turnaround time of the process.
$TAT = CT - AT$

3.3.5 Waiting Time (WT)

The time spent waiting for the CPU to complete the execution is called as waiting time of the process.
$WT = TAT - BT$

3.3.6 Response Time (RT)

The time difference between the first response and arrival times of a process is called the response time of the process.

3.4 Gantt Chart

Gantt chart shows the execution time period of the processes. For example:

P_1		P_2	P_3	
0	10	12	17	23

- Process P_1 started execution at time t = 0 and finished just before time t = 10.
- The CPU was idle from t = 10 to t = 12.
- Process P_2 started execution at time t = 12 and finished at time t = 17.
- Process P_3 started execution at time t = 17 and finished at time t = 23.

3.5 Scheduling Algorithms

3.5.1 FCFS

- It is a non pre-emptive algorithm.
- Processes are assigned to the CPU based on their arrival times. If two or more processes have the same arrival times, then the processes are assigned based on their process ids.
- It is free from starvation.
- It suffers from ***Convoy effect.***

3.5.2 Shortest Job First (SJF)

- It is a non-pre-emptive algorithm.
- Processes are assigned to the CPU based on the shortest burst time. If two or more processes have the same burst times, then the processes are assigned to the CPU based on their arrival times.
- If all the processes have the same burst times, SJF behaves as FCFS.
- In SJF, the burst times of all the processes should be known prior execution.
- It minimizes the average response time of the processes.
- There is a chance of starvation.

3.5.3 Shortest Remaining Time First (SRTF)

- It is a pre-emptive algorithm.
- Processes are assigned to the CPU based on their shortest remaining burst times.
- If all the processes have the same arrival times, SRTF behaves as SJF.
- It minimizes the average turnaround time of the processes.
- If shorter processes keep on arriving, then the longer processes may starve.

3.5.4 Longest Remaining Time First (LRTF)

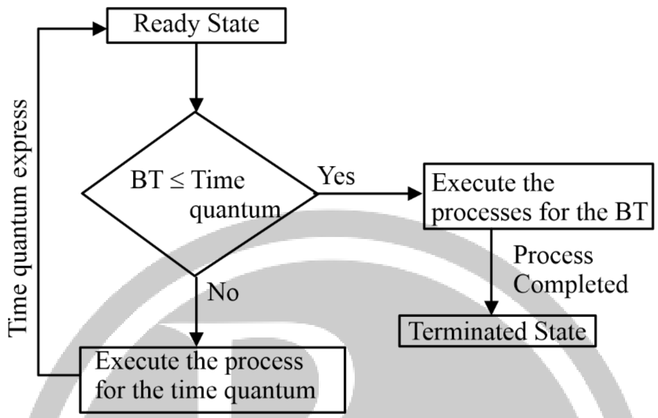
- It is a pre-emptive algorithm.
- Processes are assigned to the CPU based on their longest remaining burst times.
- It minimizes the average response time of the processes.
- If favours CPU bound processes.
- It is free from starvation.

3.5.5 Priority based Scheduling

- Priority scheduling can either be pre-emptive or non-pre-emptive.
- In pre-emptive priority scheduling, a process is voluntarily pre-empted whenever a higher priority process arrives.
- In non-pre-emptive priority scheduling, the scheduler picks up the highest priority process.
- If all the processes have equal priority, then priority scheduling behaves as FCFS.

3.5.6 Round Robin Scheduling

- RR scheduling is a pre-emptive FCFS based on the concept of time quantum or time slice.



- If the time quantum is too small, then the number of context switches (overhead) will increase and the average response time will decrease.
- If the time quantum is too large, then the number of context switches (overhead) will decrease and the average response time will increase.
- If the time quantum is greater than the burst times of all the processes, RR scheduling behaves as FCFS.

3.5.7 Highest Response Ratio Scheduling

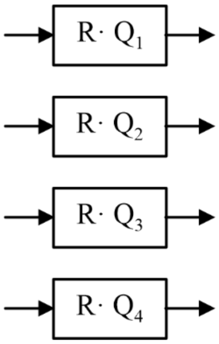
- It is a non pre-emptive algorithm.
- Processes are assigned to the CPU based on their highest response ratio.
- Response ratio is calculated as-

$$\text{Response Ratio} = \frac{WT + BT}{BT}$$

- It is free from starvation.
- It favours the shorter jobs and limits the waiting time of the longer jobs.

3.5.8 Multilevel Queue Scheduling

Multi-Level Queue Scheduling



- Depending on priority of the process, in which particular ready queue, the process has to be placed will be decided.
- High priority processes will be placed in top-level ready queue and low priority process will be placed in bottom level ready queue.
- Only after completion of all the processes, from top level ready queue the further level ready queue processes will be scheduled.
- If this is the strategy followed, then the processes which are placed in bottom level ready queue suffer from starvation.

If higher priority processes keep on arriving, the lower priority processes may starve. This problem can be solved with the help of aging. Aging is a technique which automatically increases the priority of the processes that have been waiting in the system for a very long time.